

AI Engineer Technical Assessment

Task:

Build a **Multi-Agent Research Assistant** capable of generating a comprehensive research report on a given topic.

The system should accept a research topic as input and coordinate multiple AI agents to:

1. Gather relevant information.
2. Summarize findings.
3. Verify factual accuracy.
4. Generate a final structured report.

Example Input:

"Impact of Generative AI on Software Development"

Expected Output:

A structured report containing:

- Executive Summary
 - Key Findings
 - Supporting Evidence
 - Fact Verification Notes
 - References
 - Conclusion
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Required Agents

1. Research Agent

Responsibilities:

- Search and collect information from reliable sources.
- Extract relevant content.

- Identify key insights and trends.
- Gather supporting references.

Expected Output:

- Raw research notes
 - Source URLs
 - Important findings
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2. Summarization Agent

Responsibilities:

- Process information gathered by the Research Agent.
- Remove duplicate information.
- Generate concise summaries.
- Organize information into logical sections.

Expected Output:

- Structured summary
 - Main points
 - Important observations
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3. Fact-Checking Agent

Responsibilities:

- Validate claims generated by the Research Agent.
- Cross-reference findings using multiple sources.
- Identify potential inaccuracies.
- Flag unsupported statements.

Expected Output:

- Verified facts
 - Confidence score
 - Fact-check notes
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4. Report Generation Agent

Responsibilities:

- Generate a professional research report.
- Combine outputs from all previous agents.
- Create a readable and well-structured document.
- Include citations and references.

Expected Output:

- Final report
 - References section
 - Executive summary
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Technical Requirements

Candidates may use any one of the following frameworks:

- LangGraph
- CrewAI
- AutoGen

Backend:

- FastAPI
- Django

Vector Database (Optional):

- FAISS
 - ChromaDB
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Functional Requirements

The system should:

Input Example:

```
{  
  "topic": "Impact of Generative AI on Software Development"  
}
```

Processing

1. Trigger Research Agent.
2. Pass results to Summarization Agent.
3. Send findings to Fact-Checking Agent.
4. Generate final report.

Output

```
{  
  "topic": "...",  
  "executive_summary": "...",  
  "key_findings": [...],  
  "fact_check_results": [...],  
  "references": [...],  
  "conclusion": "..."  
}
```

Deliverables

Candidates must submit:

Source Code

GitHub Repository containing:

- Complete source code
- Agent workflows
- Configuration files

Documentation

README.md containing:

- Setup instructions
- Architecture overview
- Agent workflow explanation
- Assumptions made

API Documentation

Include:

- Endpoints
- Request examples
- Response examples

Sample Output

Provide at least 2 research reports generated by the system.